MODULE *problem3*

EXTENDS *TLC*, *FiniteSets*, *Integers*

VARIABLES *boatman*, *left*, *right*

$Items \triangleq \{\text{“tiger”}, \text{“goat”}, \text{“hay”}\}$

$Init \triangleq$
$\quad \land\ boatman = \text{“left”}$
$\quad \land\ left = Items$
$\quad \land\ right = \{\}$

$TypeOK \triangleq$
$\quad \land\ \ boatman \in \{\text{“left”}, \text{“right”}\}$
$\quad \land\ \ left \in \text{SUBSET } Items$
$\quad \land\ \ right \in \text{SUBSET } Items$

$BanksOK \triangleq$
$\quad \land\ left \cap right = \{\}$
$\quad \land\ left \cup right = Items$

$Dangerous(bank) \triangleq$
$\quad \lor \{\text{“tiger”}, \text{“goat”}\} \subseteq bank$
$\quad \lor \{\text{“hay”}, \text{“goat”}\} \subseteq bank$

$SafeBank(bank, side) \triangleq$
$\quad \neg(Dangerous(bank) \land boatman \neq side)$

$Safe \triangleq$
$\quad \land\ SafeBank(left, \text{“left”})$
$\quad \land\ SafeBank(right, \text{“right”})$

$Cross(start, start_side, end, end_side, cargo) \triangleq$
$\quad \land\ boatman = start_side$
$\quad \land\ boatman' = end_side$
$\quad \land\ cargo \subseteq start$
$\quad \land\ Cardinality(cargo) \leq 1$
$\quad \land\ start' = start \setminus cargo$
$\quad \land\ end' = end \cup cargo$
$\quad \land\ Safe'$

$Next \triangleq$
$\quad \lor \exists\, cargo \in \text{SUBSET } left :$
$\qquad Cross(left, \text{“left”}, right, \text{“right”}, cargo)$
$\quad \lor \exists\, cargo \in \text{SUBSET } right :$
$\qquad Cross(right, \text{“right”}, left, \text{“left”}, cargo)$

$Goal \triangleq$
$\quad \land\ boatman = \text{“right”}$
$\quad \land\ right = Items$
$\quad \land\ left = \{\}$

$NotGoal \triangleq \neg Goal$